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T. C. Mumme, S. H. Lamb & R. I. Walcott

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The Raukumara paleomagnetic domain: constraints on the tectonic rotation of the East Coast, North Island, New Zealand, from paleomagnetic data

T. C. MUMME
Geophysics Division
Department of Scientific and Industrial Research
P. O. Box 1320
Wellington, New Zealand

S. H. LAMB*
R. I. WALCOTT
Research School of Earth Sciences
Victoria University of Wellington
P. O. Box 600
Wellington, New Zealand

*Present address: Department of Geological Sciences, University of Oxford, Oxford, United Kingdom.

Abstract New paleomagnetic data from Miocene (Altonian–Otaian) sediments from the northeastern part of the Raukumara Peninsula gave declinations close to those expected for the Australian plate (c. 20° for 20 Ma rocks), and imply that no net tectonic rotation has affected this region since that time, in contrast to the large declinations and large clockwise tectonic rotations previously recorded for the Gisborne–Wairoa region. The boundary between these two regions is constrained to lie within a broad 50 km zone between Tokomaru Bay and Gisborne.

Keywords paleomagnetism; Miocene; sediments; plate tectonics; rotation; Raukumara Domain; Wairoa Domain; East Coast; North Island

INTRODUCTION

Paleomagnetic results from the East Coast region, reported in Walcott & Mumme (1982) and Mumme & Walcott (1985), showed that a zone of structural discontinuity must exist somewhere between Hicks Bay and Gisborne to account for the large differences in the declinations of primary magnetisation for rocks of Miocene age from the two regions. The studies showed that, while the Gisborne region had undergone a clockwise rotation of some 3°–4°/Ma during the Neogene, there was no rotation for Hicks Bay during the same period. Additional paleomagnetic studies in the Gisborne region (Wright & Walcott 1986) have shown that rate of clockwise rotation was not constant throughout the Neogene but gradually increased from 2°/Ma for the period 20–25 Ma to 7°/Ma in the last 5 Ma.

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In an effort to locate more precisely the zone of structural discontinuity, and establish whether it occurs in a narrow band between the two regions or consists of a gradual increase in rotation occurring within the crust with increasing latitude, 12 localities of Miocene sediments between Hicks Bay and Tokomaru Bay were sampled for paleomagnetic studies. In addition to determining the directions of primary remanent magnetisation, measurements were made to determine the susceptibility anisotropy which can be used to categorise the nature of magnetic fabric in these sediments. Recent geological mapping by Victoria University of Wellington and New Zealand Geological Survey was used in the selection of outcrops sampled (Fig. 1).

For this study, ages of 16.3, 18, and 22 Ma are assigned to the boundaries of the Clifdenian–Altonian, Altonian–Otaian, and Otaian–Watakian Stages. These values are different from those published by Hoskins (1982) but reflect recent unpublished revisions (Hoskins pers. comm. 1988).

Sampling

At the chosen outcrops, 25 mm diameter oriented cores were drilled at regular intervals (usually every 2–3 m). Since the sedimentation rates for Tertiary sediments from the East Coast are estimated at about 0.5 mm/year (Walcott & Mumme 1982; Wright 1986), cores were collected over a stratigraphic thickness of at least 10 m, which correlates to a time interval of at least 20 000 years, and is therefore sufficient to average out secular variation effects (McElhinney 1973). At each outcrop the finer grained horizons were more frequently sampled as these tended to give more stable paleomagnetic directions. Below is a brief summary of the localities sampled.

HK4 — Hicks Bay Cores 2504–2554; 2668–2679. Map reference NZMS 260 Z14/718877–712870. About 300 m of Whakai Formation (Otaian Stage) outcrops in a stream on the south side of State Highway 35 about 1 km west of Hoia Station and midway between localities HK2 and HK3 (Walcott & Mumme 1982) (Fig. 1a). The section is a fine, blue-grey mudstone at the stratigraphic top, becoming gradually coarser grained towards the base. Cores 2504–2554 were collected in groups of 2 or 3 at average intervals of 10 m down the section. Paleomagnetic measurements on these cores indicated a polarity change midway down the section and a subsequent collection of cores 2668–2679 was drilled every 2–3 m over this apparent transition zone from normal to reversed polarity. Details of the nature of this transition are the subject of further work. Strike and dip of the outcrop was 083°T/26°N at the top and 083°T/37°N at the base of the section. (T indicates the bearing refers to true north.)
AGE: G. H. Scott, New Zealand Geological Survey (pers comm.), assigned ages based on foraminifera identified in seven samples collected at regular intervals down this section as Otaian Stage, and therefore an age of c. 20 Ma is likely.

TA1 — Karakatuwhero River Cores 2569–2580. Map reference NZMS 260 Z14/747777. Outcrop of siltstone/sandstone in a sidestream on the right bank of the Karakatuwhero River and about 800 m upstream from the Karakatuwhero valley road-end (Fig. 1b). The base of the section sampled is about 13 m stratigraphically above the unconformity with the lower Tertiary mudstone. Cores

were collected upsection at about 3 m intervals. Average strike and dip of the outcrop was 180°T/37°E.

TA2 — Karakatuwhero River Cores 2581–2590. Map reference NZMS 260 Z14/749782. Outcrop of fine-grained mudstone in a small sidestream on the right bank of Karakatuwhero River about 250 m upstream from the Karakatuwhero valley road-end (Fig. 1b). Cores were collected at about 3 m intervals over 26 m of the section. The locality is about 100 m stratigraphically above TA1. Average strike and dip of the outcrop was 166°T/18°E.
AGE: NZGS fossil locality Z14/f7017 (about 400 m stratigraphically above our locality) is Otaian. NZGS fossil localities Z14/f6602, 3 are in the same position as TA1 (above), and the samples were recorded as collected just above the unconformity between the Paleogene and Neogene sediments. Therefore the sediments at TA2 are probably mid Otaian, and an age of c. 20 Ma is assigned.

TA3—Ngararahiwihiwi Stream Cores 2591–2603. Map reference NZMS 260 Z15/786685. Outcrop of mudstone with thin, black sandstone beds in Ngararahiwihiwi Stream and about 3 km upstream from where it joins the Poroporo River (Fig. 1d). The cores were collected at 1 m intervals from a stratigraphically continuous sequence which is repeated about every 10 m by low-angle normal faulting. Average strike and dip of the outcrop was 196°T/28°E.
AGE: No fossil age determinations have been done on samples from Ngararahiwihiwi Stream; however the similarity of rock type with the Altonian rocks from locality TA7, about 3 km to the south, probably indicates a similar age (i.e., c. 17 Ma).

TA4—Ngararahiwihiwi Stream Cores 2604–2621. Map reference NZMS 260 Z15/784687. Outcrop of blue-grey mudstone/siltstone in Ngararahiwihiwi Stream about 100 m upstream from TA3 and just upstream from a prominent bend in the stream (Fig. 1d). Cores were collected at regular intervals over a stratigraphic thickness of 40 m. Average strike and dip of the outcrop was 206°T/19°E.
AGE: About 17 Ma (see TA3).

TA5—Whakatu Stream Cores 2622–2638. Map reference NZMS 260 Z15/764563. Outcrop of blue-grey mudstone/siltstone with local sandstone beds in Whakatu Stream about 1500 m upstream from where it crosses State Highway 35 (Fig. 1e). Cores were collected upsection at regular intervals over a stratigraphic thickness of 57 m. Average strike and dip of the outcrop was 066°T/14°NW.

TA6 — Tangakaka Stream Cores 2639–2667. Map reference NZMS 260 Z14/798729–800728. Outcrop of blue-grey mudstone/siltstone with many tuffaceous beds in Tangakaka Stream and about 2.5 km upstream from where it crosses the Whakaangiangi–Taurangakautuku road (Fig. 1c). Cores were collected at regular intervals over a stratigraphic thickness of 200 m. Some tuffs were also sampled. Strike and dip of the outcrop was 283°T/70°S at the base and 279°T/41°S at the top of the section sampled.
AGE: NZGS fossil locality Z14/f35, situated in Tangakaka Stream and midway along the section sampled, is identified as at the base of Altonian Stage, and therefore an age of c. 18 Ma is assigned.

TA7—Poroporo River Cores 2680–2695. Map reference NZMS 260 Z15/795655. Outcrop of siltstone in a small sidestream on the left side of the Poroporo River and about 500 m west of Waihuku Station farmhouse (Fig. 1d). Cores were collected at about 2 m intervals over a stratigraphic thickness of 21 m. Average strike and dip of the outcrop was 342°T/31°SW.
AGE: NZGS fossil locality Z14/f7592 (map reference NZMS 260 Z15/795656) is identified as Altonian Stage, and therefore an age of c. 17 Ma is assigned to the section we sampled.

WR1—Waiau River Core 2555–2568. Map reference NZMS 260 Y16/648260. Outcrop of fine-grained siltstone in a roadcut on a forestry road about 300 m west of the Makomako Station road (Fig. 1f). Average strike and dip of the outcrop was 353°T/18°W.

WR2 — Tironui Station, Waiau River Cores 2696–2702. Map reference NZMS 260 Y16/689250. Outcrop of coarse-grained siltstone on the left bank of the Waiau River and about 1 km downhill from the Tironui Station road-end (Fig. 1f). Cores were collected at regular intervals over a stratigraphic thickness of 9 m. Average strike and dip of the outcrop was 346°T/20°W.

WR3—Makomako Station Bridge, Waiau River Cores 2703–2714. Map reference NZMS 260 Y16/654258. Outcrop of fine-grained siltstone in a roadcut by the Makomako Station Bridge (Fig. 1f). Cores were collected at regular intervals over a stratigraphic thickness of 21 m. Average strike and dip of the outcrop was 007°T/20°W.

MA1 — Mangaroa Stream Cores 2769–2797. Map reference NZMS 260 Y16/681198. Outcrop of blue-grey mudstone in the Mangaroa Stream, adjacent to the Mangaroa Station farmhouse and where the bridge on the Waiau Road crosses stream (Fig. 1g). Cores were collected at regular intervals over a stratigraphic thickness of 20 m. Average strike and dip of the outcrop was 025°T/28°W.
AGE: NZGS fossil locality Y16/f569 (map reference NZMS 260 Y16/681198) is identified as Altonian Stage, and therefore an age of c. 17 Ma is assigned to the section we sampled.

Magnetic measurements

In the laboratory, 22 mm length samples were cut from the cores and their natural remanent magnetisation (NRM) measured on a MOLSPIN spinner magnetometer. Selected samples from each locality were then either stepwise demagnetised in alternating fields (A.F.) at 5 mT increments, or thermally demagnetised at temperatures of 100, 200, 250, 300, 350, 400, 500, and 600°C, to determine optimum fields/temperatures at which to demagnetise the remaining samples. The A.F. methods proved more effective in eliminating the secondary components of magnetisation. Optimum demagnetising fields (i.e., where the primary magnetisations were clearly recognisable) were generally between 15 and 30 mT, and the remaining samples were demagnetised at these levels.

A.F. demagnetisation equipment used was either a MOLSPIN A.F. tumbling demagnetiser or a SCHONSTEDT 3 axis A.F. demagnetiser. Magnetometers used for the measurements of remanence on the main batch of samples were a MOLSPIN spinner magnetometer, a DIGICO spinner magnetometer and a ScT cryogenic magnetometer. The magnetic susceptibility anisotropy of selected samples from several localities was measured using a DIGICO anisotropy delineator at Centre des Faibles Radioactivités, Gif-sur-Yvette, France.

RESULTS

Saturation magnetisation

Saturation isothermal remanence experiments carried out on two samples from each locality indicated that saturation was always achieved in low fields, <0.2 T (Fig. 2), showing that hematite was not the principal magnetic carrier but that the carrier is likely to be one of the minerals from the titanomagnetite series (Tarling 1983). The saturation magnetisations (M_s) for samples from the more northern localities (i.e., TA2, TA4, TA7 and HK4) were much greater than those from the Tokomaru Bay region (i.e., WR2 and WR3). This is probably related to the proximity of the source rocks.

Susceptibility anisotropy

A measurement of the degree of deformation that a sediment has undergone can be gained from the determination of magnetic susceptibility anisotropy. Volume magnetic susceptibility anisotropy is the property of the rock in which

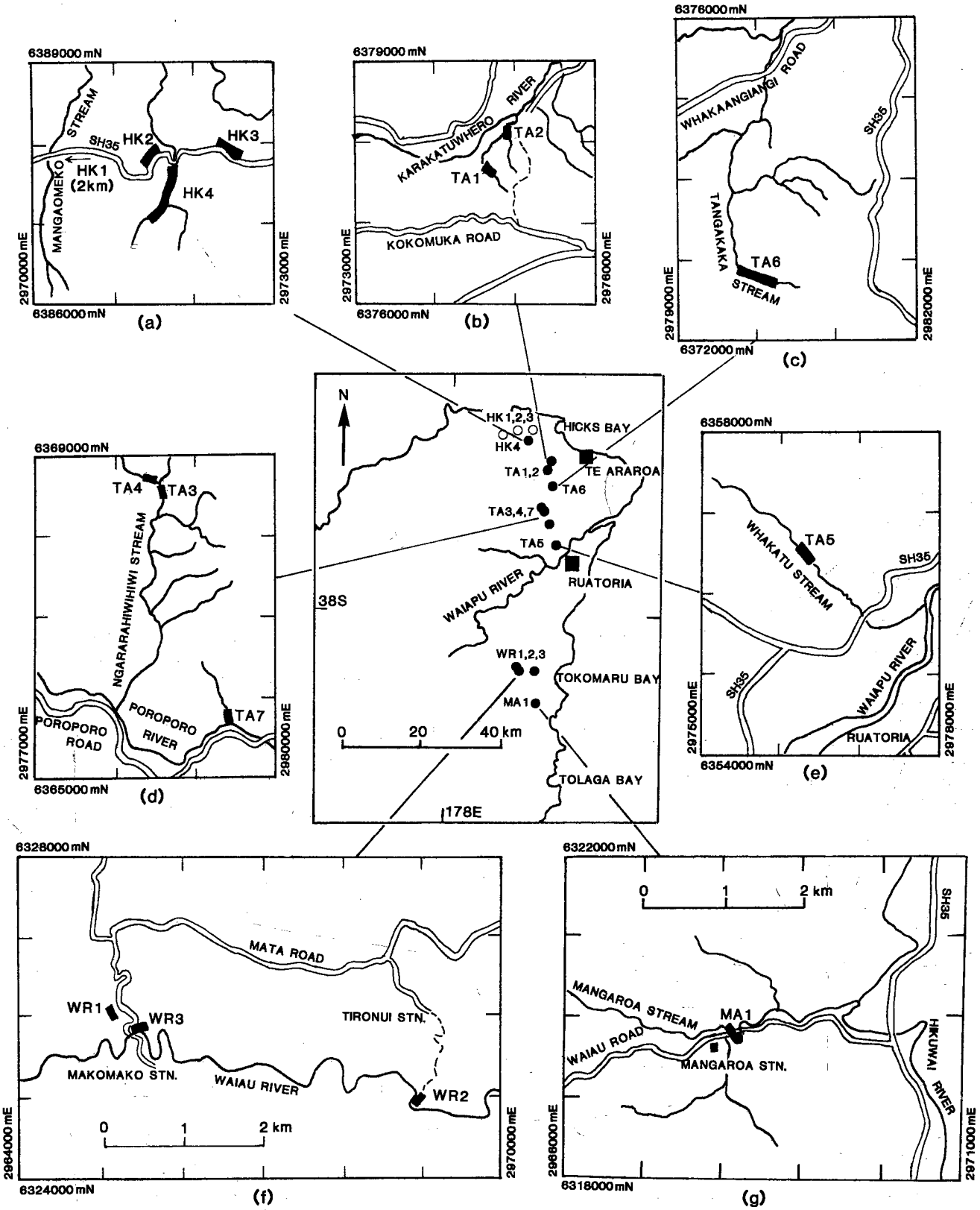


Fig. 1 Localities from where cores have been collected for paleomagnetic studies. Solid circles, this study. Open circles, Walcott & Mumme (1982) and Mumme & Walcott (1985). Grid co-ordinates are referenced to the NZ metric map grid.

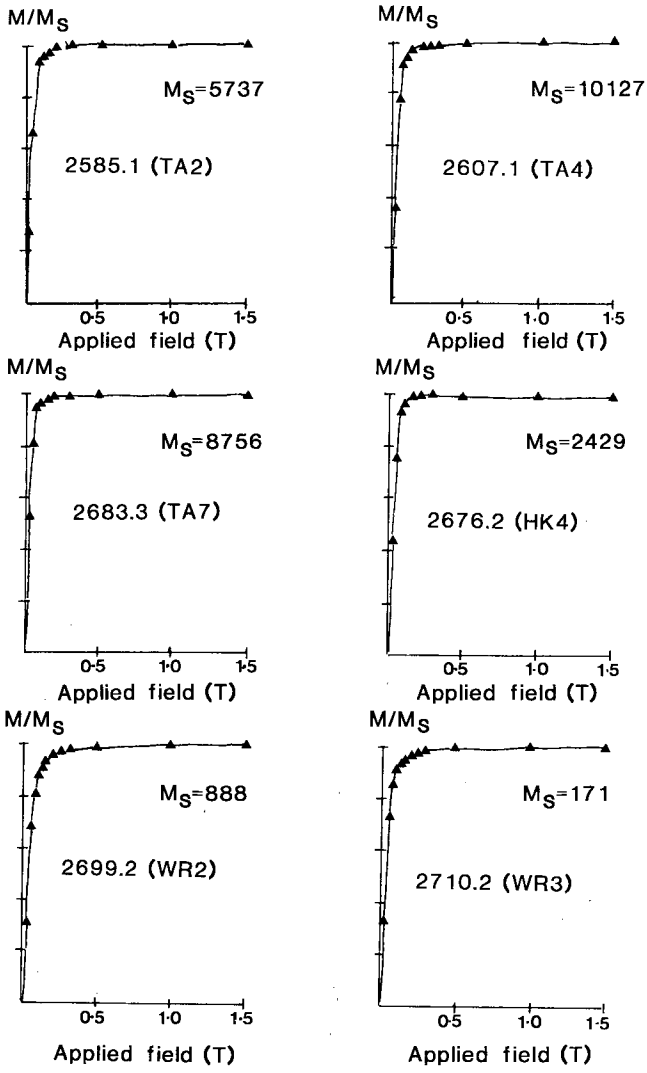


Fig. 2 Acquisition of isothermal remanence for selected samples (localities in brackets). Saturation magnetisation (M_s) is in units of 10^{-3}A/m.

the magnitude of the magnetic susceptibility varies with direction. It can be approximated by a symmetric tensor of second rank characterised by three principal susceptibilities k_1, k_2, k_3, with relative magnitudes $k_1 \geq k_2 \geq k_3$. Geometrically such a tensor can be visualised as a triaxial ellipsoid with semi-axes, proportional to k_1, k_2, k_3. When $k_1 > k_2 \approx k_3$ the ellipsoid is a prolate spheroid; when $k_1 \approx k_2 > k_3$, it is an oblate spheroid. The usual assumption of susceptibility isotropy would be given by $k_1 = k_2 = k_3$ for which the spheroid reduces to a sphere. Many combinations of these principal susceptibilities have been proposed to characterise the magnetic fabric of a rock (Tarling 1983, table 6.3). These parameters include: lineation = k_1/k_2, foliation = k_2/k_3, degree of anisotropy = k_1/k_3, prolateness = $(k_1 - k_2)/(k_2 - k_3)$, and oblateness = $(k_2 - k_3)/k_1 - k_2)$. In addition, the mean volume susceptibility is given as $k_{mean} = (k_1 + k_2 + k_3)/3$. The relationship between magnetic fabric and visible petrofabric of a rock is a complicated one and not completely understood (Kligfield et al. 1981).

In undeformed sediments the magnetic susceptibility is characterised by an oblate spheroid with the minimum susceptibility axis, k_3, normal to the bedding plane, and any observed magnetic lineation has generally been considered as resulting from the presence of depositional currents (Hamilton & Rees 1970; Hrouda 1982). The susceptibility anisotropy studies on sediments from the East Coast localities indicate a consistent pattern of k_3 axes normal to the bedding plane (Fig. 3), and the ratios of $k_1/k_2 < k_2/k_3$ indicate that the ellipsoid of anisotropy is strongly oblate (Fig. 4). These two results are consistent with the planar sedimentary fabric observed from the East Coast sediments. Some lineation was evident with the k_1 axes clustering along the east–west direction of the bedding plane, and this is probably attributed to the paleocurrent directions. This lineation was more evident in some localities (i.e., TA3, TA4, TA7, and HK4) where the predominant lithology was siltstone. The lithology at TA2 is a fine-grained mudstone deposited in a quiet environment where the paleocurrent was much less strong and hence less lineation is evident. The larger scatter observed in the k_1 and k_3 axes at localities WR2 and 3 is probably due to the weaker

Table 1 Mean magnetic susceptibilities and ratios and their standard errors for selected samples from seven localities on the East Coast.

Locality	*n*	k_1	k_2	k_3	k_{mean}
HK4	12	1.64 ± 0.16	1.60 ± 0.16	1.48 ± 0.15	1.57 ± 0.16
TA2	7	0.92 ± 0.05	0.92 ± 0.05	0.84 ± 0.05	0.89 ± 0.05
TA3	8	1.25 ± 0.08	1.23 ± 0.08	1.13 ± 0.07	1.20 ± 0.08
TA4	15	1.10 ± 0.03	1.07 ± 0.03	1.01 ± 0.03	1.06 ± 0.03
TA7	8	1.10 ± 0.04	1.07 ± 0.04	1.01 ± 0.04	1.06 ± 0.04
WR2	4	0.21 ± 0.01	0.20 ± 0.01	0.20 ± 0.01	0.20 ± 0.01
WR3	5	0.18 ± 0.01	0.18 ± 0.01	0.17 ± 0.01	0.18 ± 0.01

	$\frac{100(k_1-k_3)}{k_1}$	k_1/k_3	k_1/k_2	k_2/k_3	$\frac{(k_2-k_3)}{(k_1-k_2)}$	$\frac{(k_1-k_2)}{(k_2-k_3)}$
HK4	9.75 ± 0.33	1.108 ± 0.004	1.030 ± 0.002	1.076 ± 0.004	2.54 ± 0.41	0.45 ± 0.03
TA2	9.48 ± 0.85	1.105 ± 0.011	1.010 ± 0.001	1.094 ± 0.010	9.36 ± 1.50	0.13 ± 0.03
TA3	9.49 ± 0.24	1.105 ± 0.003	1.018 ± 0.002	1.086 ± 0.003	4.91 ± 0.62	0.23 ± 0.03
TA4	7.48 ± 0.48	1.081 ± 0.006	1.020 ± 0.002	1.061 ± 0.006	3.46 ± 0.51	0.41 ± 0.07
TA7	8.06 ± 0.78	1.088 ± 0.010	1.024 ± 0.005	1.063 ± 0.005	3.34 ± 0.68	0.38 ± 0.06
WK2	4.78 ± 0.25	1.050 ± 0.003	1.006 ± 0.001	1.044 ± 0.003	7.49 ± 1.30	0.15 ± 0.03
WK3	4.18 ± 0.44	1.044 ± 0.005	1.007 ± 0.001	1.037 ± 0.004	5.28 ± 0.54	0.20 ± 0.02

n, number of selected samples; k_1, k_2, k_3, principal axes of susceptibility ellipsoid (units of 10^{-3} S.I.); k_{mean}, mean volume magnetic susceptibility (units of 10^{-3} S.I.); $100(k_1-k_3)/k_1$, percentage anisotropy; k_1/k_3, degree of anisotropy; k_1/k_2, lineation; k_2/k_3, foliation; $(k_2-k_3)/(k_1-k_2)$, oblateness; $(k_1-k_2)/(k_2-k_3)$, prolateness.

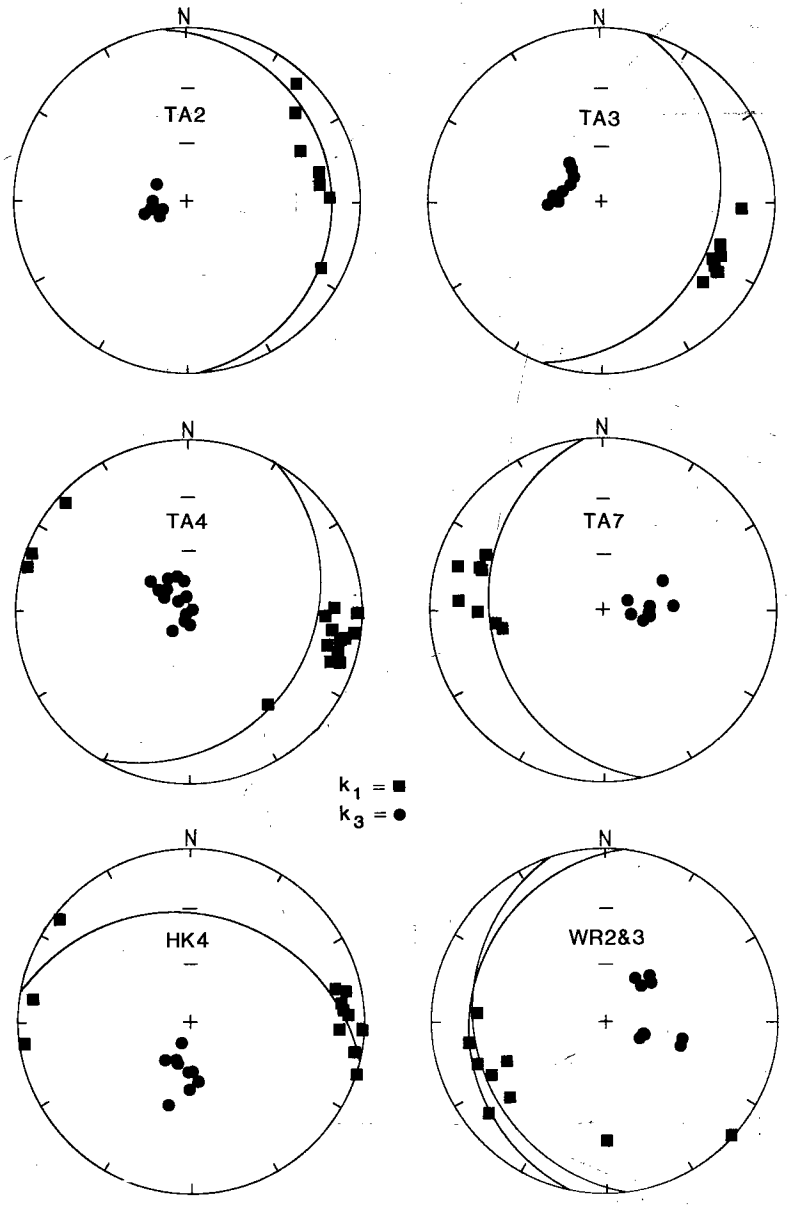


Fig. 3 Stereographic lower hemisphere projections of the directions of the k_1 and k_3 susceptibilities (before bedding correction) for samples from localities on the East Coast. The bedding plane is projected and the k_3 susceptibilities are clearly normal to the bedding plane. The clustering of the k_1 susceptibilities indicate that some lineation exists.

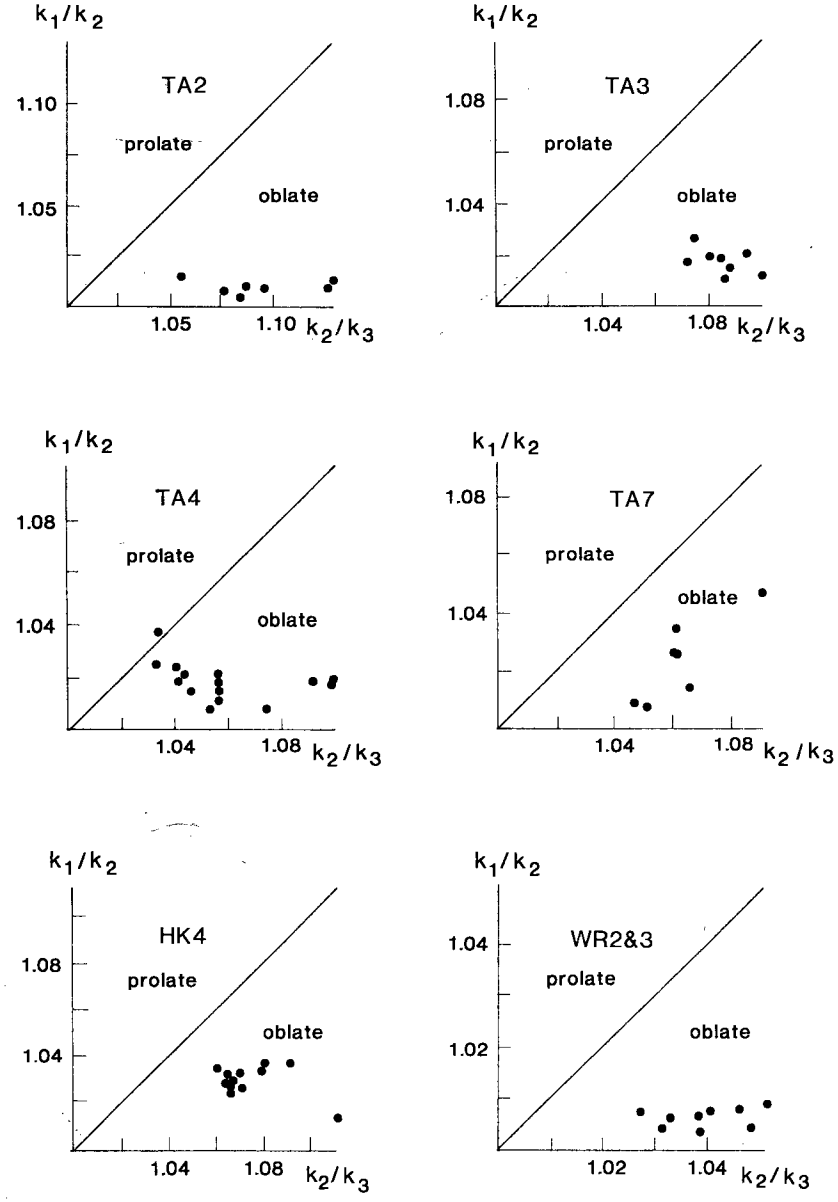


Fig. 4 Anisotropy plots of k_1/k_2 (lineation) against k_2/k_3 (foliation) for samples from localities on the East Coast. The susceptibility ellipsoids are oblate for most samples and thus indicate a planar sedimentary fabric.

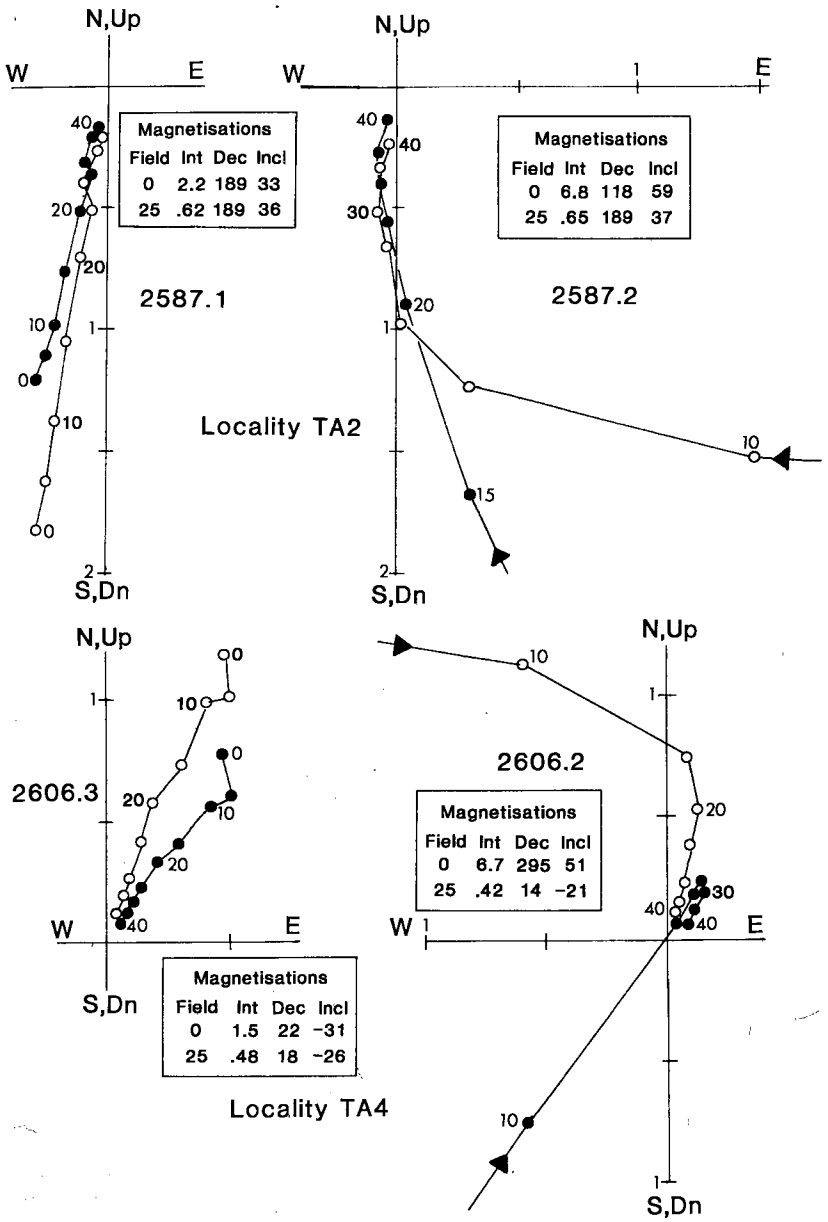


Fig. 5 Zijderveld plots for samples from reversed and normal polarity rocks illustrating demagnetisation behaviour for adjacent uncontaminated (2587.1 and 2606.3) and contaminated (2587.2 and 2606.2) samples. The laboratory-acquired viscous magnetisation was eliminated after demagnetisation in fields of 20–25 mT. Open circles are the projections of the magnetisation vectors onto the horizontal (N, E, S, W) plane; solid circles are the projections onto the vertical (up, down, E, W) plane. Magnetisations are in units of 10^{-2}A/m; demagnetisation fields in mT.

mean susceptibilities and the smaller mean degree of anisotropy (k_1/k_3) (Table 1).

Kissel et al. (1986) observed in consolidated marine sediments in the Aegean that, with increasing deformation, a prolate or "pencil" structure develops about which the k_2 and k_3 axes form a girdle. The direction defined by the clustering of the k_1 axes is normal to the maximum compressive strain. Kissel et al. (1986) interpreted this as a tectonic fabric. The direction of present-day, maximum, relative compressional strain for two regions in the East Coast has been geodetically observed at 54° ± 9 and 48° ± 9 (Walcott 1984). Whereas for some localities the k_1 axes are normal to these directions (e.g., TA3) it is not universally so, and the k_2 and k_3 axes do not form a girdle around the direction of the k_1 axes. Therefore, although these sediments are relatively consolidated and tilted through low angles, the deformation is not sufficient to obscure the sedimentary fabric. Such sediments are ideal for paleomagnetic studies.

Table 2 Changes in total intensity (M) and (Z) component of magnetisation during demagnetisation for selected samples. Intensities in units of 10^{-3} A/m.

		NRM		10 mT		20 mT		30 mT	
Locality	Sample	M	Z/M	M	Z/M	M	Z/M	M	Z/M
TA1	*2571.3	94.6	0.96	14.7	0.86	3.6	–0.32	2.7	–0.67
	2571.4	14.9	–0.82	9.5	–0.79	5.0	–0.82	2.1	–0.80
TA2	*2587.2	68.2	0.97	20.6	0.76	8.6	0.54	4.4	0.46
	2587.1	22.0	0.42	17.0	0.45	8.8	0.46	5.0	0.44
TA3	*2591.3	110.8	0.87	34.3	0.88	12.7	0.75	5.6	0.72
	2591.1	34.7	0.65	30.0	0.70	16.4	0.70	8.5	0.72
TA4	*2606.3	66.7	0.97	14.6	0.70	5.7	–0.32	2.7	–0.52
	2606.2	15.4	–0.55	11.5	–0.50	7.0	–0.49	3.0	–0.41
TA7	*2685.3	66.2	0.99	9.8	0.85	3.0	–0.38	1.6	–0.51
	2685.1	13.2	–0.78	6.4	–0.76	4.5	–0.70	1.9	–0.73

*The dominant component of NRM was along the Z axis of these samples implying contamination by a laboratory-acquired viscous remanent magnetisation.

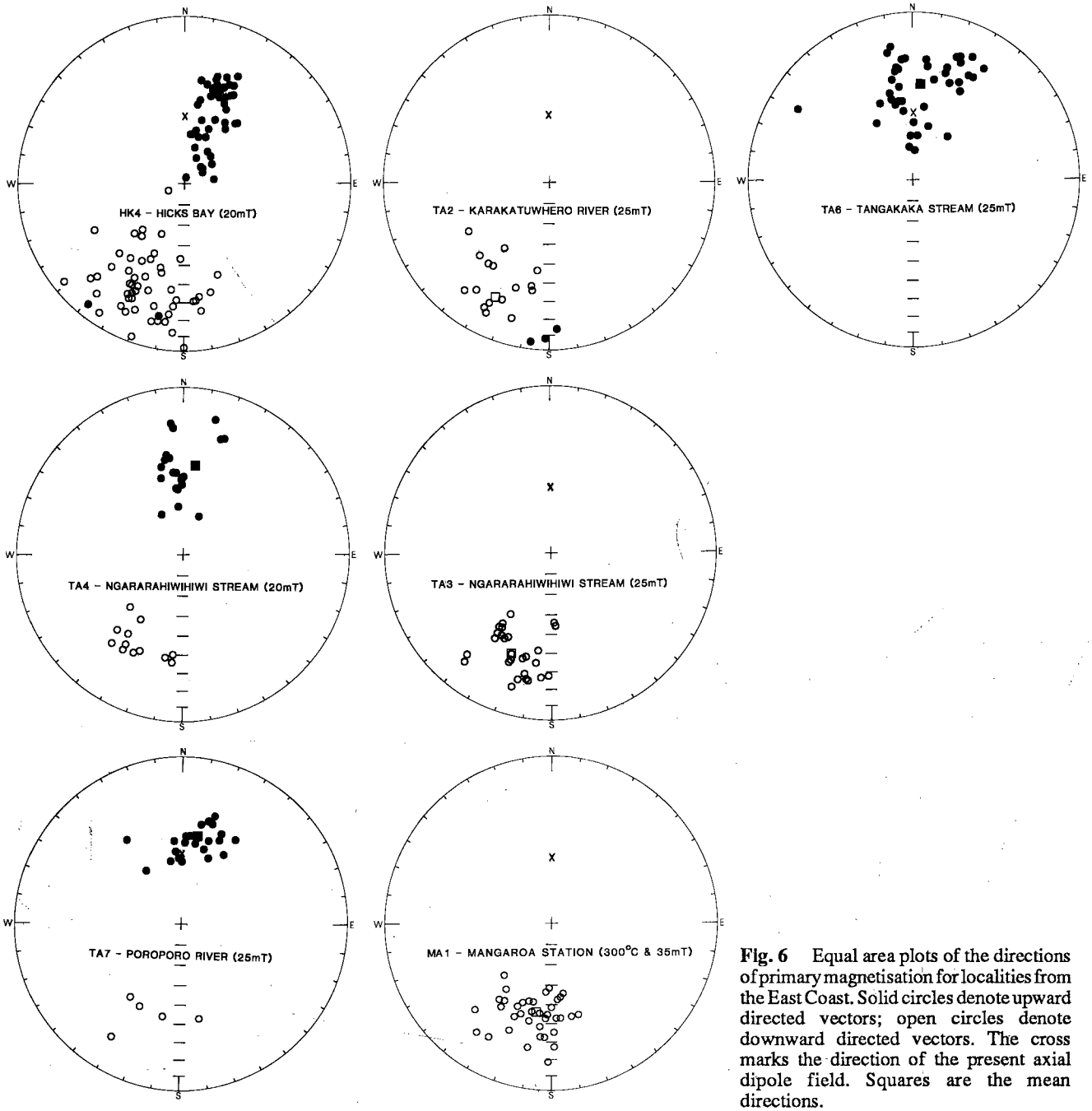


Fig. 6 Equal area plots of the directions of primary magnetisation for localities from the East Coast. Solid circles denote upward directed vectors; open circles denote downward directed vectors. The cross marks the direction of the present axial dipole field. Squares are the mean directions.

Contamination by secondary magnetisation

(1) Laboratory-acquired viscous remanent magnetisation

A number of samples from each locality acquired a secondary viscous remanent magnetisation (VRM) characterised by NRM intensities very much greater than the intensities of adjacent samples (Table 2). We assume that this VRM was acquired in the laboratory because the dominant component of NRM was always directed along the *Z* axis of the sample regardless of the orientation of the core (Table 2). Walcott & Mumme (1982) and Mumme & Walcott (1985) observed a similar laboratory-acquired VRM in samples collected from Tertiary sediments, and concluded that this was acquired during storage, as it only appeared in those stored without the protection of magnetic shielding. The laboratory-acquired VRM is easily removed in alternating fields of < 30 mT. Its appearance in some samples and not in others from the same core remains unexplained, however, as all samples were subjected to the same storage treatments between collection and measurement. Table 2 shows changes in intensities on demagnetisation for contaminated and uncontaminated samples from some localities. Zijderveld plots (Fig. 5) illustrate this demagnetisation behaviour for adjacent samples from two localities on the East Coast.

(2) Secondary magnetisation in the present-day field direction

Samples from 5 of the 12 localities (TA1, TA5, WR1, WR2 and WR3) were contaminated by a hard secondary magnetic overprint in the direction of the present-day magnetic field ($D = 20, I = -63$). The groupings of the NRM directions, before tilt correction, for samples from these localities were excellent with $\alpha_{95} < 3.5°$ but all were close to the present-day magnetic

field direction implying a secondary overprint (Table 3). Attempts to remove this secondary overprint by either A.F. or thermal demagnetisation methods resulted in a breakdown of stability of magnetisation for samples from these localities.

Transition zones

At three localities (TA4, TA7 and HK4) we sampled through apparent transition zones between reversed and normal polarity. Exact lengths of the zones were not determined due to the sparsity of the sampling interval. However, on average, each transition spans a thickness of 5–10 m of section. Sedimentation rates for Tertiary sediments from these areas are estimated at about 0.5 mm/year (Wright 1986; Walcott & Mumme 1982) thereby implying an average time of transition from one polarity to the other of between 10^4 and 2×10^4 years.

Primary magnetisation

Primary magnetisation was recognised in samples from 7 of the 12 localities and the results are summarised in Table 4 and Fig. 6. At these localities samples were considered to possess

Table 3 Mean NRM directions (before tilt correction) for five localities.

Locality	n	D	I	α_{95}
TA1	33	15	–61	3.26
TA5	37	18	–56	2.71
WR1	25	1	–55	3.48
WR2	15	17	–61	3.41
WR3	22	13	–58	3.28
Present-day magnetic field direction		20	–63	

n, number of samples; D, declination of remanent magnetisation; I, inclination of remanent magnetisation (-ve = upward directed vectors); α_{95}, radius of the 95% confidence circle about the mean.

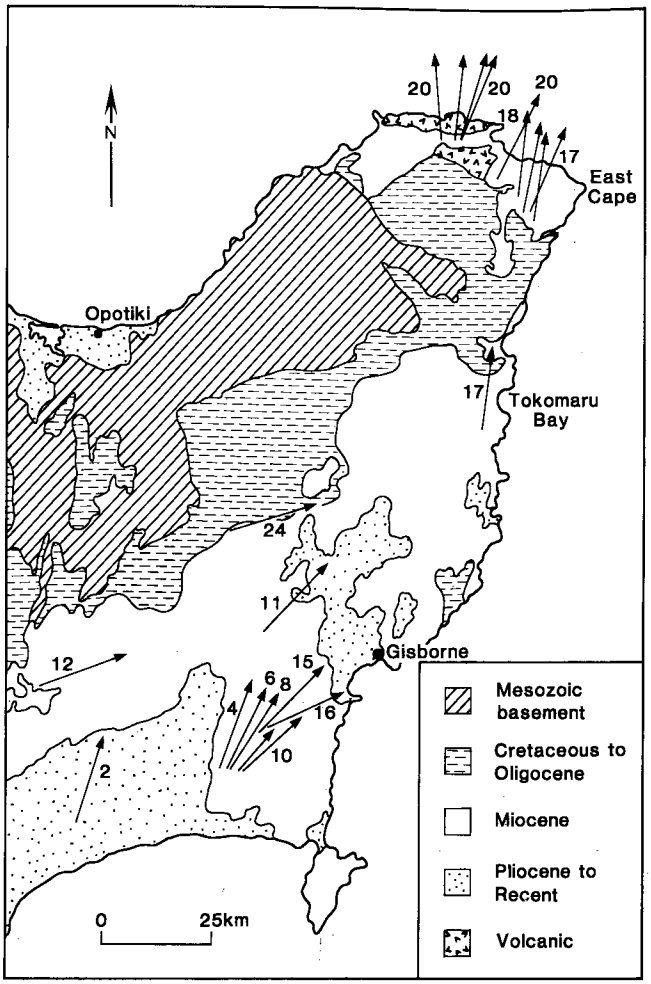


Fig. 7 Declinations for localities in the East Coast region. Approximate ages (in Ma) are shown beside the arrows. The difference in declinations of the Tokomaru Bay–Hicks Bay and Gisborne–Wairoa groups indicate that a zone of structural discontinuity must exist between Tokomaru Bay and Gisborne.

Table 4 Summary of new paleomagnetic data from the Raukumara Domain.

Locality		Cores	Age* (Ma)	n	D	I	α_{95}	$d \pm \delta$
TA2	(R)	2581–2590	20 ± 1	19	203	25	8.94	23 ± 8
TA3	(R)	2591–2603	17 ± 1	27	201	36	4.95	21 ± 5
TA4	(N)	2604–2610	17 ± 1	21	359	–46	6.52	–1 ± 8
	(R)	2617–2621		12	208	40	6.27	28 ± 7
	(All)	2604–2621		33	9	–45	5.66	9 ± 6
TA6	(N)	2643–2667	18 ± 1	42	4	–42	5.65	4 ± 6
TA7	(R)	2680–2681	17 ± 1	5	200	40	13.61	20 ± 14
	(N)	2684–2695		23	8	–48	4.89	8 ± 6
	(All)	2680–2695		28	10	–47	4.76	10 ± 6
HK4	(N)	2504–2520	20 ± 1	44	24	–53	4.55	24 ± 6
	(R)	2521–2544		54	203	32	5.92	23 ± 6
	(All)	2504–2544		98	22	–42	4.12	22 ± 4
MA1	(R)	2769–2797	17 ± 1	35	189	44	4.02	9 ± 4

*Ages assigned using data obtained from NZGS fossil catalogue (see text).
n, number of accepted samples. Samples not included in the analysis were those collected from transition zones (i.e., at localities TA4, TA7, HK4) and those having no recognisable primary magnetisation.
D, declination of remanent magnetisation.
I, inclination of remanent magnetisation (+ve, downward directed; -ve; upward directed)
α_{95}, 95% cone of confidence about the mean direction.
d, clockwise rotation from the present axial dipole.
δ, 95% confidence limit in declination rotation = $0.8[\sin^{-1}(\sin \alpha_{95}/\cos I)]$.

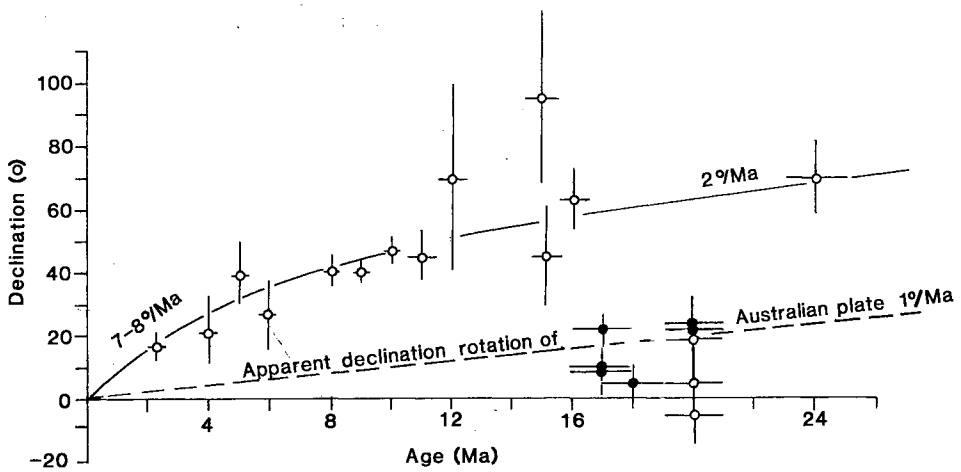


Fig. 8 Plot of age versus declination and associated errors for the East Coast paleomagnetic data. Solid circles represent new paleomagnetic data presented in this paper and open circles are data previously reported (see text). The data from the northeastern part of the Raukumara Peninsula cluster close to the curve of apparent declination rotation of Australian plate 1°/Ma and imply that this region has not undergone any tectonic rotation since the Early Miocene.

a recognisable primary magnetisation only if they exhibited a directional stability of magnetisation through the 20, 25, and 30 mT demagnetisation levels, and had either reversed directions or normal directions (before tilt correction) that were significantly different from the direction of the present-day magnetic field.

Shallow inclinations were recorded at all localities (Table 4). The expected inclination for 20 Ma rocks from the East Coast is 60° whereas for our localities the mean reversed inclinations are +35°; mean normal inclinations are –47°. This discrepancy may occur either during deposition by the alignment of the magnetic grains parallel to the sedimentation surface or during consolidation and compaction. Although early laboratory experimental studies (King 1955) have shown the existence of the former, it has rarely been observed in natural sediments. It is now generally considered that postdepositional realignment of the small, single-domain magnetic grains in the direction of the ambient field occurs (McElhinney 1973; Verosub 1977; Tarling 1982). The shallowing of the inclinations during compaction have been observed in sediments (Noltimier & Richardson 1980) and this is the most likely cause for the shallow inclinations observed here.

Regardless of the causes of these apparent inclination errors the declination recorded in sediments is the magnetic parameter least influenced by depositional, postdepositional, and compactional processes and should reliably record the magnetic field at the time the rock was formed.

DISCUSSION

The declinations of Table 4 and those obtained from Tertiary sediments in the Gisborne–Wairoa region (details of which are reported in Walcott & Mumme 1982; Mumme & Walcott 1985; Wright & Walcott 1986) are plotted by area in Fig. 7 and by age in Fig. 8. All localities north of Tokomaru Bay have declinations close to those expected for the Australian plate based on the Australian Cenozoic apparent polar path (Idnurm 1985) and therefore no tectonic rotation has evidently affected this region (Fig. 8). Localities in the Gisborne–Wairoa region, however, show large declinations and indicate very substantial rotation. Much of the rotation occurred in the last 10 Ma (Fig. 8); however, the scatter and uncertainties in older data are such that any change in declination with greater age is poorly constrained. We refer to these two regions as the **Raukumara Domain** in the northeast and the **Wairoa Domain** in the southwest. When referenced to the Pacific plate, for which Cenozoic paleomagnetic pole positions show no apparent polar wander (Grindley et al. 1977), the Raukumara Domain has rotated at about the same rate as the Australian plate, implying no differential or tectonic rotation, whereas the Wairoa Domain has rotated some 50° in the last 10 Ma (Wright & Walcott 1986).

The late Tertiary sediments of the Raukumara Domain are extensively cut by active normal faults, widely observed in outcrop and well described by Mazengarb (1984). Lamb (1988) gave a map of part of the Raukumara Domain showing extensive normal faulting that has occurred. A shear-strain rate of about 0.3×10^{-6} radians/year, with an extension axis of 140°, has been observed geodetically, consistent with the normal faulting. These features have been explained as thin-skinned, extensional faulting produced by gravitational collapse of the upper part of an accretionary prism, the collapse being due to the thickening of the prism by underplated sediments (Walcott 1987).

Normal faulting also occurs in the Wairoa Domain but is not as extensively developed, except locally (Pettinga 1982). Indeed, in this rotated block, the characteristic feature is a lack of deformation in late Tertiary sedimentary rocks, the principal structure being the 70 km wide, gently folded Wairoa Syncline.

There is no obvious structural change between the two domains that may account for their different rotational behaviour. Conceptually, the change need not be a discontinuity like a fault or fault system — it could be a change from a simple-shear regime over the Wairoa Domain with its consequent vorticity to a pure-shear irrotational regime over the Raukumara Domain. Such a model of pervasive finite shear is very difficult to accept in this region of generally gentle dips and internally undeformed sediments. We would prefer models involving the rotation of the Wairoa Domain en bloc, and any such model requires decoupling structures (faults, fault systems or zones of relatively intense extension, shear or compression) between the two domains.

The southern limit to the Raukumara Domain is presently determined to lie south of Tokomaru Bay, and the northwestern limit of the Wairoa Domain is close to Gisborne. Prospects of determining the limits more precisely are not good — more than 20 Tertiary localities between Tokomaru Bay and Gisborne were sampled for this purpose but none gave a stable magnetisation. Further collections are likely to be as unproductive.

The boundary between the two domains is determined to lie within a broad, ill-defined zone. The requirements for the boundary structure are that it permits rotation of the Wairoa Domain a minimum of 30° without affecting the Raukumara Domain. A possible structure could be a wrench fault between Gisborne and Opotiki (Fig. 7) forming a circular arc with centre about half-way between Opotiki and East Cape. A total slip of about 40 km with a present-day rate of 5 mm/year would totally decouple the two domains. But many other structures are conceivable. The identification of the nature of this boundary remains one of the important problems to be solved so as to fully understand the paleomagnetic information from the East Coast of the North Island.

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